

Abstract No.

Modeling SHH medulloblastoma and treatment-induced neurodevelopmental alterations using Human cerebellar organoids

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Sonic Hedgehog (SHH)–type medulloblastomas are embryonal tumors that disrupt cerebellar cortex development. Although survival rates reach 75–88%, many survivors experience long-term cognitive and emotional impairments. Chemotherapy, radiotherapy treatments contribute to these sequelae, highlighting the need to understand their underlying mechanisms on both healthy and tumoral tissues. Loss of function of PTCH1, a key SHH pathway receptor, promotes abnormal proliferation of cerebellar progenitors and tumorigenesis.

We generated cerebellar organoids from wild-type human induced pluripotent stem cells (hiPSCs) and hiPSCs carrying heterozygous PTCH1 inactivation (PTCH1+/-). The effects of SHH-targeted therapies were assessed using Vismodegib or JQ1, administered either acutely (day 28) or chronically (28, 35 and 42). Cerebellar and SHH pathway gene expression was analyzed by RTqPCR. Major cell markers, as well as primary cilia bearing the Ptch1 receptor, were analyzed by immunohistochemistry. Spontaneous electrical activity was recorded in mature organoids using microelectrode arrays, and the involvement of GABAergic and glutamatergic receptors was investigated.

In healthy organoids, Vismodegib inhibited SHH signaling, as shown by complete loss of GLI1 expression, and unexpectedly induced a ciliopathy characterized by predominantly biciliated cells. JQ1 exacerbated this phenotype, leading to multiciliated cells. Organoids developed functional neuronal networks involving GABA-A receptors, while Vismodegib reduced spontaneous activity. PTCH1+/- organoids displayed increased growth but preserved structural and electrophysiological features. Following Vismodegib treatment, they similarly showed complete GLI1 loss and multiciliation.

Recently, we established a new invasion model by introducing SHH-group medulloblastoma cells into healthy cerebellar organoids, providing a physiologically relevant system to investigate tumor–cerebellar interactions. We aim to use this model to further assess how anticancer therapies affect both medulloblastoma invasion and the surrounding healthy cerebellar tissue.

Overall, cerebellar organoids reveal how SHH-targeted therapies disrupt ciliogenesis and neuronal activity, providing a human-relevant platform to decipher treatment-induced neurodevelopmental defects and refine therapeutic strategies in SHH medulloblastoma.



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